Improved $\dot{V}O_2$max and Time Trial Performance With More High Aerobic Intensity Interval Training and Reduced Training Volume: A Case Study on an Elite National Cyclist

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Abstract

Støren, Ø, Bratland-Sanda, S, Haave, M, and Helgerud, J. Improved $\dot{V}O_2$max and time trial performance with more high aerobic intensity interval training and reduced training volume: a case study on an elite national cyclist. *J Strength Cond Res* 26(10): 2705–2711, 2012—The present study investigated to what extent more high aerobic intensity interval training (HAIT) and reduced training volume would influence maximal oxygen uptake ($\dot{V}O_2$max) and time trial (TT) performance in an elite national cyclist in the preseason period. The cyclist was tested for $\dot{V}O_2$max, cycling economy (C_c), and TT performance on an ergometer cycle during 1 year. Training was continuously logged using heart rate monitor during the entire period. Total monthly training volume was reduced in the 2011 preseason compared with the 2010 preseason, and 2 HAIT blocks (14 sessions in 9 days and 15 sessions in 10 days) were performed as running. Between the HAIT blocks, 3 HAIT sessions per week were performed as cycling. From November 2010 to February 2011, the cyclist reduced total average monthly training volume by 18% and cycling training volume by 60%. The amount of training at 90–95% HRpeak increased by 41%. $\dot{V}O_2$max increased by 10.3% on ergometer cycle. TT performance improved by 14.9%. C_c did not change. In conclusion, preseason reduced total training volume but increased amount of HAIT improved $\dot{V}O_2$max and TT performance without any changes in C_c. These improvements on cycling appeared despite that the HAIT blocks were performed as running. Reduced training time, and training transfer from running into improved cycling form, may be beneficial for cyclists living in cold climate areas.

Key Words road cycling, endurance, 1-year period

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Introduction

Time trial (TT) cycling on high national or international level is a typical aerobic endurance event, as it is a single start race against time and lasts from approximately 40 to 70 minutes. Aerobic energy supply dominates the total energy requirements after about 75 seconds of near-maximal exercise (5). After 120 minutes, the aerobic/anaerobic distribution is approximately 99/1% (23). This implies that less than 5% of the metabolic energy release during TT is anaerobic. Time performance in typical aerobic endurance events is according to di Prampero (4), physiologically limited by $\dot{V}O_2$max, the distance covered and the fractional utilization of $\dot{V}O_2$max over that distance and oxygen cost of moving (C).

Støa et al. (21) have shown fractional utilization of $\dot{V}O_2$max to be negligible for time performance when the duration of competition is less than 20 minutes but becoming increasingly more important, as the duration of the competition increases beyond 30 minutes, as shown by Davies and Thompson (3). In endurance sports at high performance level, such as running (21) and cross-country skiing (11), there have been reported a very strong relationship between $\dot{V}O_2$max and race performance level. In cycling, however, Lucia et al. (16) report that a good cycling economy expressed by a low oxygen cost of cycling (C_C) seems to compensate for relatively low $\dot{V}O_2$max values among world-class professional road cyclists. According to Lucia et al. (15), C_C and LT are more important determinants of endurance cycling performance than $\dot{V}O_2$max. They speculate that it is the huge cycling volume among the professional cyclists that gives them a favorable C_C.

A high training volume with a high percentage of easy to moderate training intensity is in accordance with a typical example presented in Seiler and Kjerland (20) regarding junior cross-country skiers, where 75% of the total training volume averaged 65% $\dot{V}O_2$max. This type of training is also currently the recommendation from the Norwegian Olympic Training Sports Centre for athletes in Norwegian national teams in endurance sports.

TABLE 1. Time course of testing and training.

Time and season	February 2010	Race season	November 2010	Postseason training	November 2010	Postseason training	January 2011	Preseason training	February 2011	Preseason training	February 2011
Training		Traditional training		HAIT block (running)		Reduced total training volume and cycling volume		HAIT block (running)		Reduced total training volume and cycling volume	
		High total volume, high cycling volume		14 HAIT sessions in 9 days		3 HAIT sessions per week (both running and cycling)		15 HAIT sessions in 10 days		3 HAIT sessions per week (both running and cycling)	
Testing	$\dot{V}O_2max$ cycling LT C_C, TT, anthropometry		$\dot{V}O_2max$ cycling $\dot{V}O_2max$ running		$\dot{V}O_2max$ cycling $\dot{V}O_2max$ running		$\dot{V}O_2max$ cycling $\dot{V}O_2max$ running		$\dot{V}O_2max$ cycling $\dot{V}O_2max$ running		$\dot{V}O_2max$ cycling LT C_C, TT, anthropometry

Despite high volume models, studies on patients (19), healthy students (8), soccer players (7,17), and top level marathoners (1) have shown high aerobic intensity interval training (HAIT) at 85–95% $\dot{V}O_2max$ to be superior to moderate training in increasing $\dot{V}O_2max$. Studies using HAIT sessions among cyclists have found 4.9–8.1% improvement in $\dot{V}O_2max$ after training periods with 2–3 HAIT sessions per week for 3 or more weeks (6,12–14). The improvement in $\dot{V}O_2max$ was correlated with improvements in TT in Laursen et al. (14).

Some studies indicate that block periodization of HAIT followed by sufficient recovery can result in improvements in $\dot{V}O_2max$ without the athletes showing symptoms of overtraining syndrome. Overtraining syndrome is defined as a status of imbalance between training and recovery, and this status does not improve after 14 days of rest (22). Breil et al. (2) found 7.5% improvements of $\dot{V}O_2max$ in alpine skiers after 15 HAIT sessions within 11 days but no report of overtraining syndrome. To our knowledge, the effects of such extreme blocks of HAIT in cyclists have not yet been reported.

One of the basic principles of training is the principle of training specificity. For cyclists, this means as much of the endurance training as possible on a cycle. This specificity principle is also in accordance with findings that $\dot{V}O_2max$ results in athletes are modified by type of activity performed during the test. In a literature review by Millet et al. (18), $\dot{V}O_2max$ among runners and cyclists was shown specific to exercise modality. However, larger physiological training transfers were found from running to cycling than vice versa (18).

If $\dot{V}O_2max$, fractional utilization of $\dot{V}O_2max$, and C (4) are of equal importance for TT performance, the cyclist should theoretically complete a high weekly amount of cycling, both to comply with the principle of training specificity and to improve C_C and fractional utilization of $\dot{V}O_2max$ (15). $\dot{V}O_2max$ may however play a more important role for TT performance than C_C and fractional utilization of $\dot{V}O_2max$ (11,14,21). In this case, a reduced training volume to allow for more HAIT training may enhance TT performance (2,6,12–14) even in an elite cyclist. Because cardiac output seems to be closely linked to $\dot{V}O_2max$ (8), all of HAIT training may not even be performed as cycling as long as the training leads to at least the same stress on the cardiac system. In countries with a cold winter climate, changing some of the HAIT training from cycling to running during winter preseason would also give practical advantages to the cyclist.

High-level cyclists already have high $\dot{V}O_2max$, lactate threshold, and good C_C, and therefore, it is interesting to examine the effects of blocks of HAIT sessions on a national elite-level cyclist. The aim of this study was thus to examine to what extent increased volume of HAIT with a reduced total training volume program during preseason would influence $\dot{V}O_2max$ and TT performance in a national elite-level male cyclist. A secondary aim of the study was to

TABLE 2. Training during the 1-year intervention period.*†

	February to October	November to January	Δ (min)	Δ (%)
Mean monthly total training volume (min)	3,494	3,154	−340	−10
Mean monthly training volume, cycling (min)	3,250	1,290	−1,960	−60
Mean monthly training volume, running (min)	244	1,864	1,620	764
Mean monthly training volume at 60–80% HRmax (min)	3,112	2,488	−624	−20
Mean monthly training volume at 81–89% HRmax min)	222	164	−58	−26
Mean monthly training volume at 90–95% HRmax (min)	150	212	62	41
Mean monthly training volume at 96–100% HRmax (min)	11	4	7	−34

*HR = heart frequency ($b \cdot min^{-1}$).
†Values are minutes and percentages presented as mean per month and differences between the periods February to October and November to January.

evaluate if HAIT performed as running would give benefits on cycling $\dot{V}O_2$max and cycling performance.

Our hypotheses were as follows:

1. Reduced total training volume and increased duration of HAIT improves $\dot{V}O_2$max.
2. HAIT sessions performed as running improves $\dot{V}O_2$max in cycling.
3. Improved $\dot{V}O_2$max improves TT cycling performance.

METHODS

Experimental Approach to the Problem

The main objective of the present study was to investigate to what extent more HAIT and reduced training volume would improve $\dot{V}O_2$max and TT performance in an elite national cyclist in the preseason period. To do this, the cyclist was tested for $\dot{V}O_2$max, cycling economy (C_c), and TT performance on an ergometer cycle during 1 year. Training was continuously logged using heart rate (HR) monitor during the entire period. Total monthly training volume in the 2011 preseason was reduced compared with the 2010 preseason, and 2 HAIT blocks (14 sessions in 9 days and 15 sessions in 10 days) were performed as running during preseason training to the 2011 season. Between the HAIT blocks, 3 HAIT sessions per week were performed as cycling. Being a case study, this design may give an answer to whether it is possible even for an elite cyclist to improve $\dot{V}O_2$max and TT performance by reducing total training volume, increasing HAIT volume, and change some of the cycle training into running.

Subjects

A male national elite road cyclist, specialized in TT races, age 24 years, 86 kg (body weight), and 190 cm (height) was followed during 1 year (2010–2011). The year 2010 was the third season at top national level for this athlete. The athlete was informed of the experimental risks and signed an informed consent document before the investigation. The investigation was approved by the institutional review board. It was also approved by the Regional Ethical Committee of southern Norway. Subject characteristics are presented in Table 1. During this year, all training was logged using Garmin Edge 500 HR monitor and Trainingpeaks WKO+ version 3.0 software program for the analyses of the HR files (Peaksware, Lafayette, CO, USA). The 1-year period started preseason in February 2010. There were no major differences in the preseason training for 2010 compared with the prior preseason training (Table 2).

Procedures

At the start of the period, the cyclist was tested for $\dot{V}O_2$max, C_C, LT, percent body fat (on test, day 1), and a TT (on test, day 2). Percent body fat was measured with a skin caliper (Lange; Beta Technology, Santa Cruz, CA, USA) at the chest, the thigh, the suprailiac, the abdomen, and the triceps brachii. The cycling tests were all carried out using an electrically braked cycle (Lode Excalibur Sport; Lode, Groningen, the Netherlands). All $\dot{V}O_2$ measurements were performed using the metabolic test system SensorMedics Vmax Spectra 229 (SensorMedics, Yorba Linda, CA, USA). The athlete was instructed always to use his freely chosen cadence during testing.

The LT and C_C assessment was started at 150 W, assumed to represent 30–40% of $\dot{V}O_2$max, to assure an easy warm-up value. This was held for 5 minutes. The brake power was then increased to 200 W for the next 5 minutes and then to 250 W. Every 5 minutes from there on, the brake power was increased by either 10 or 25 W until the protocol terminated at just above LT, which was defined as the lowest blood lactate concentration ($[La^-]_{blood}$) + 2.3 $mMol \cdot L^{-1}$. $[La^-]_{blood}$ was measured using the Arkray Lactate Pro LT-1710 (Arkray, Inc., Kyoto, Japan). This protocol is based on the protocol described by Helgerud et al. (9). C_C was calculated at the brake power output representing 70% of $\dot{V}O_2$max. After 10 minutes of easy cycling at 150 W, the $\dot{V}O_2$max test started.

TABLE 3. Test results form February 2010 to February 2011.*†

	Time of testing and results					
	February 2010	November 2010	November 2010	January 2011	February 2011	Δ February 2010 to February 2011 (%)
Anthropometry						
BW	87.6	85.5	85.6	86.0	85.9	−1.7 (1.9)
% body fat	13.4				13.5	0.1 (0.7)
TT (cycling)						
Time (s)	1,698				1,445	−253 (14.9)
Power (W)	329.4				378.5	49.1 (14.9)
%$\dot{V}O_2max$	79.1				77.5	−1.6 (2.1)
R	0.98				1.0	0.02 (2.0)
$[La^-]_{blood}$	5.8				6.8	1.00 (17.2)
Cadence (RPM)	89.8				95.3	5.5 (6.1)
HR	179.0				172.6	−6.4 (3.6)
$\dot{V}O_2max$ (running)						
l·min		5.50	5.73	5.86	6.02	
$ml·kg^{-1}·min^{-1}$		64.3	67.1	68.6	70.0	
$ml·kg^{-0.67}·min^{-1}$		279.2	290.7	297.3	304.7	
HR		194	185	188	186	
R		1.12	1.10	1.04	1.09	
$\dot{V}O_2max$ (cycling)						
l min	5.83	5.53	5.98	6.35	6.34	0.51 (8.7)
$ml·kg^{-1}·min^{-1}$	66.6	64.7	70.0	73.9	73.6	7.0 (10.5)
$ml·kg^{-0.67}·min^{-1}$	291.4	280.7	303.5	321.3	320.1	28.7 (10.3)
HR	193	193	185	185	189	−4 (2.1)
$[La^-]_{blood}$	11.8	10.8	10.4	10.3	13.6	1.8 (15.2)
R	1.08	1.06	1.07	1.06	1.10	0.02 (1.9)
W	485	480	500	540	560	75 (15.5)
C_C (cycling)						
$ml·kg^{-1}·W^{-1}$	0.157				0.153	−0.004 (2.5)
$ml·kg^{-0.67}·W^{-1}$	0.678				0.669	−0.009 (1.3)
LT (cycling)						
%$\dot{V}O_2max$	76.2				80.2	4.0 (5.2)
W	333.3				380.0	46.7 (14.0)

*BW = body weight; HR = heart rate; $[La^-]_{blood}$ = blood lactate concentration; power = $Nm·s^{-1}$; R = respiratory exchange ratio; RPM = rounds per minute; $\dot{V}O_2max$ = maximal oxygen uptake; %$\dot{V}O_2max$ = fractional utilization.

†Values are test results, differences in results from February 2010 to February 2011, and percent difference in parentheses.

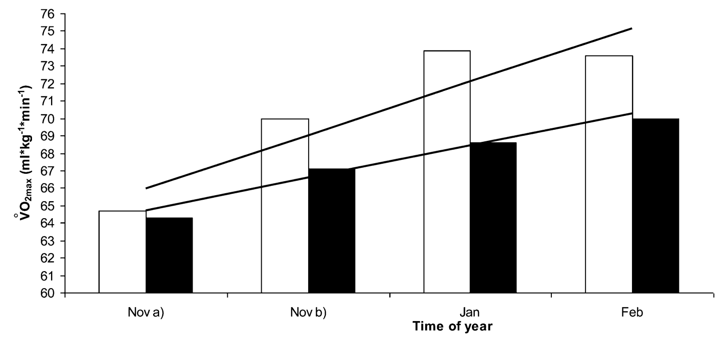

Figure 1. Maximal oxygen consumption from November 2010 to February 2011, $\dot{V}O_2$max, maximal oxygen consumption in ml·kg^{-1}·min^{-1}. Open bars represent cycling, and closed bars represent running. Upper linear trend line represents cycling, and lower linear trend line represents running. Nov = November; Jan = January; Feb = February.

The brake power was set to 100 W below the wattage at lactate threshold (LT_W). During the first minute, the brake power was gradually increased up to LT_W. From there on, the power was increased by 10–25 W every 30 seconds, based on the subjective evaluation of the test leader. At voluntary fatigue, the test terminated. In addition, ≥95% HRmax, R >1.05, $[La^-]_{blood}$ >8.0 mMol·L^{-1}, and a plateau of the $\dot{V}O_2$ curve were the criteria for $\dot{V}O_2$max (23). For running, the test protocols were the same as described above, only with velocity instead of brake power on a 5% inclined treadmill (Woodway PPS 55 Sport, Waukesha, Germany). The run test was performed first, followed by 60 minutes of easy cycling at 150 W before the cycle test. Heart rate was measured continuously during the tests, and $[La^-]_{blood}$ was measured immediately after the tests.

Time trial was tested on the second day of testing after a 20-minute warm-up. The test was performed over 15 km on the test cycle. This corresponds approximately to 23 km on a flat outdoor road, due to an underestimation by the test bikes flywheel of approximately 50% of the covered distance. This result is based on pilot studies at our laboratory. $\dot{V}O_2$ and HR were continuously measured during TT. Immediately after the test, $[La^-]_{blood}$ was measured. Heart rate was measured during all tests using Polar s610 HR monitors (Polar Electro Oy, Kempele, Finland). All tests were performed between 10 and 11:55 AM each day. The pretest meals were standardized, as were the nutritional and fluid intake between the single tests.

Experimental Protocol

The cyclist continued the 2010 season training and competition as usual (Table 3). Training for the 2011 season started in November 2010, and 2 blocks of HAIT running sessions were performed during the 2011 preseason training. These blocks were performed in November 2010 (14 HAIT sessions in 9 days) and in January 2011 (15 HAIT sessions in 10 days). When 2 HAIT sessions were performed the same day, the cyclist always had from 8 to 10 hours of rest between sessions. Between these 2 blocks, 3 HAIT sessions per week were performed using cycling. Figure 1 shows the distribution of the HAIT sessions during the 2 blocks.

Each HAIT session during the HAIT blocks consisted of uphill running on a treadmill, 4 repetitions of 4 minutes at 90–95% of HRmax. Between each 4-minute run, there was 3 minutes of jogging at 70% of HRmax. The HAIT sessions performed between the 2 blocks (i.e., the period from medio November 2010 to medio January 2011) were performed as cycling. The athlete did not receive nutritional counseling.

Statistical Analyses

As this is a 1-year case study, all training and test data are presented only by descriptive statistics.

Results

Monthly average training volume from February 2010 to October 2010 was 3,495 minutes. Ninety-three percent of this training was performed as cycling, and 4% of the training was at the intensity between 90 and 95% of HRmax. From November 2010 to February 2011, monthly average training volume was 2,868 minutes. Forty-five percent of this training was performed as cycling, and 7% of the training was at the intensity between 90 and 95% of HRmax. The cyclist did not report any illness or injury or signs of overtraining as defined by Urhausen (22) during the training period from November 2010 to February 2011.

Body weight and percent body fat remained the same during the 1-year period. $\dot{V}O_2$max improved by 10.5%, from 66.6 to 73.6 ml·kg^{-1}·min^{-1}, and the ergometer's TT performance improved by 14.9%.

The fractional utilization of $\dot{V}O_2$max during the TT remained unchanged. Heart rate during TT decreased by 6.4%, whereas cadence (rounds per minute) increased by 5.5%. Theoretical maximal aerobic power (MAP), calculated by $\dot{V}O_2max/C_C$, improved from 424 to 481 W (13.5%). LT_W improved by 14% from February 2010 to February 2011.

Discussion

The main finding in the present study is that TT performance improved by 14.9%, which is close to the 13.5% improvement in theoretical MAP. The largest improvement affecting MAP is by far the 10.3% improvement in $\dot{V}O_2max$.

The 2 blocks of HAIT consist of 29 HAIT sessions. This gives an average improvement of 0.35% per session. However, $\dot{V}O_2max$ actually decreased by 3.8% from February 2010 to October 2010. The real improvement in $\dot{V}O_2max$ in the block period is therefore 14%, which is an average improvement of 0.5% per session. This is in line with studies using HAIT 2–3 times per week for 8–10 weeks. Helgerud et al. (8) improved $\dot{V}O_2max$ by 0.3% per session (healthy students), and McMillan et al. (17) improved $\dot{V}O_2max$ by 0.5% per session (soccer players). The present results are also in line with results from other HAIT block studies like Breil et al. (2), which improved $\dot{V}O_2max$ by 0.5% per session (alpine skiers).

In a case study, test reliability is of crucial importance. Variations in test-to-test measurements from a single person will affect the total results more than in a group of subjects where 2-tailed statistical tests may be used. $\dot{V}O_2$ measurements with the SensorMedics Vmax Spectra are from the producer reported to be accurate within a range of $\pm$3%. However, test-to-test variations with the Vmax Spectra in our laboratory are shown to be less than $\pm$1%, with a standard error mean of 0.1–0.2 in different tests, as reported in Helgerud et al. (10). The margin of error in wattage accuracy for Lode Excalibur Sport test ergometer bike is from the producer reported to be $\pm$2% between 100 and 1,500 W. The Lactate Pro lactate analyzer is reported from the producer to operate within a 3% coefficient of variance. When calculating LT, a model described by Helgerud et al. (9) was used in the present study. This model calculates LT as the warm-up $[La^-]_{blood}$ + 1.5 $mMol \cdot L^{-1}$ (unhemolyzed whole blood measured with the YSI lactate analyzer [YSI, Yellow Springs Istruments, Yellow Springs, OH, USA]). Helgerud et al. (9) report an interindividual variation of approximately 30% (from 1.3 to 1.7 $mMol \cdot L^{-1}$). The test results from February 2010 to February 2011 in the present study thus reveal differences in TT performance, $\dot{V}O_2max$, and watt at LT well outside any margin of error discussed above. The differences in C_C, LT (%$\dot{V}O_2max$), and $[La^-]_{blood}$ measured after TT or $\dot{V}O_2max$ tests are however within the margins of error discussed above.

The finding that HAIT performed as running improved cycling $\dot{V}O_2max$ and TT performance is provocative because the subject is an elite cyclist. However, the cyclist still performed a considerable amount of cycling in the HAIT block period, although the monthly training volume performed as cycling dropped by 61% in this period compared with the period from February to October. These results indicate that even a reduced total training volume in cycling is sufficient to maintain C_C and to even improve $\dot{V}O_2max$ and TT performance, as long as $\dot{V}O_2max$ is improved by an adequate activity such as running. The beneficial effect from running on cycling is in accordance with results presented in the study by Millet et al. (18).

According to the logic of science and research, the level of knowledge in the area must guide the choice of methods and design for obtaining new information. Using this principle, we argue that a randomized controlled trial on the effects of extreme blocks of HAIT sessions on endurance elite athletes was preferable, but at this point premature. There is a risk of overtraining after a HAIT block intervention. This study showed that the cyclist tolerated these great amounts of HAIT training, although he showed clear signs of overreaching in the first 3–5 days after each of the 2 HAIT block periods. We can however not generalize these results to all elite-level cyclists. For example, it is possible that younger cyclists without this athlete's previous amounts of cycling volume will experience a deteriorated C_C with reduced cycling volume. Therefore, there is a need for further research on a larger group of cyclists. In conclusion, preseason reduced training volume and increased duration of HAIT improved $\dot{V}O_2max$ and TT performance without any changes in C_c. These improvements on cycling appeared although the HAIT blocks were performed as running.

Practical Applications

Based on the results in this single case study, there are indications that blocks of increased duration of HAIT combined with reduced total training volume can enhance performance and aerobic capacity in elite endurance athletes. The improvement in cycling, although the HAIT sessions were performed as running, indicates a beneficial effect for cyclists living in cold climate areas with snowy and icy roads, as parts of the winter training may be performed indoor on treadmill during preseason or as cross-country skiing. Athletes who want to use HAIT blocks as part of their training regimen should be particularly accurate with energy intake and make sure that they are in an energy surplus state. The results from the present study regarding HAIT blocks are based on preseason training and do not transfer to in season training. To ensure optimal recovery conditions is in our opinion crucial for obtaining these improvements.

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